

Feature Engineering: Analytical Methodology

Strategic Metrics Built from a Cross-Sectional Dataset

Analytical Constraints

The biggest analytical constraint of this project was the raw dataset: it lacked timestamps, a loyalty score, and a churn label. Because we did not have a timeseries of every single order, traditional models like Recency (from RFM analysis) or Time-to-Churn were impossible. We had to rely entirely on Frequency, Monetary Value, and behavioral snapshots to engineer strategic metrics from scratch.

Below are the five specific features engineered in Python and the strategic business logic behind each one.

1. Estimated LTV (Lifetime Value)

The Steps

- We multiplied the customer's current *Purchase Amount (USD)* by their *(Previous Purchases + 1)*.

The Logic

The dataset was cross-sectional, meaning we only had a snapshot of their current cart value. To build a retention strategy, we needed a mathematically grounded proxy to estimate their total historical spend and value to the brand.

2. Promo Dependency Score

The Steps

- We evaluated two columns: *Discount Applied* and *Promo Code Used*.
- If both were 'Yes' → score of **1.0** (Highly Dependent).
- If one was 'Yes' → score of **0.5**.
- If neither → score of **0.0** (Organic).

The Logic

The project brief required us to measure how much current revenue relies on promotions. A simple Yes/No boolean was not enough; we needed a continuous score to accurately separate true organic demand from strict bargain hunters.

3. Satisfaction Flag

The Steps

- We imputed 37 missing values in the *Review Rating* column with the median.
- Any customer with a rating of **4.0 or higher** was flagged as Satisfied (1); anything below as Unsatisfied (0).

The Logic

We needed a way to quantify product satisfaction without a direct survey. This was critical for the Promotional Sunset Plan, as we needed to find segments of customers who loved the product so much that they would likely continue buying even if we took their discounts away.

4. Customer Value Tiers

The Steps

- We ranked the entire customer base by total spend into percentiles:
- Top 10% → **Platinum** | Next 20% → **Gold** | Next 30% → **Silver** | Bottom 40% → **Bronze**

The Logic

The brand needed a structured way to segment its customers to feed the Customer Pyramid visual for the Power BI dashboard.

5. The Loyalty Score (Hypothesis Testing)

The Steps

The case study explicitly required us to build at least two competing definitions of "loyalty" and argue for the superior metric using predictive logic, not intuition.

- **Hypothesis A (Transactional):** Score based on a customer's percentile rank in Previous Purchases and Estimated LTV.
- **Hypothesis B (Behavioral):** Score based on Frequency of Purchases and Subscription Status.

The Logic & Result

We ran a correlation matrix in Python to test both scores against Estimated LTV. Hypothesis A showed a staggering **0.95 correlation**, while Hypothesis B showed a near-zero **0.01 correlation**. This mathematically proved to the founders that in their specific business, loyalty is strictly driven by raw purchase volume and spend size — not by automated subscriptions.
